

# SecurePass: Arduino-Based Password Lock System

## Project Report

Abstract: SecurePass is an Arduino-based password authentication system using a 4x4 matrix keypad and 16x2 LCD display. The system validates a user-entered password and grants or denies access.

## Introduction

Security is important in homes, offices, and industries. This project demonstrates a simple password-based electronic lock using Arduino UNO, keypad input, and LCD feedback.

## Objectives

- Interface keypad and LCD with Arduino
- Implement password authentication
- Learn Embedded C and hardware interfacing

## Components Used

Arduino UNO, 4x4 Matrix Keypad, 16x2 LCD Display, Potentiometer, Breadboard, Jumper Wires

## Working Principle

1. User enters password.
2. Arduino reads keypad input.
3. Password is compared with stored password.
4. LCD displays 'Access Granted' or 'Access Denied'.

## Applications

Home Security, Office Access Control, Electronic Door Locks, Educational Projects

## Advantages

Low cost, easy implementation, user friendly, reliable authentication.

## Future Scope

RFID, Fingerprint Sensor, IoT Connectivity, Mobile App Control, GSM Alerts.

## Conclusion

The project successfully demonstrates a password-based security system using Arduino UNO, keypad interfacing, and LCD display technology.